# MULTI-TEMPORAL WATER BODY ANALYSIS OF HYDERABAD'S URBAN LAKES USING DEEP LEARNING




**Submitted By -**

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**Under the Guidance of -**

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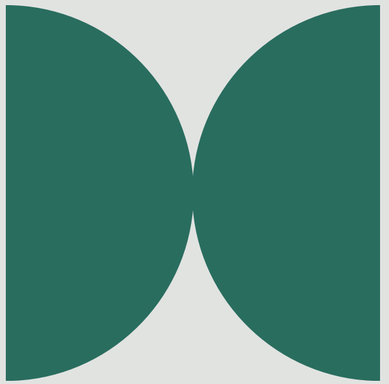
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2026

# OUTLINE

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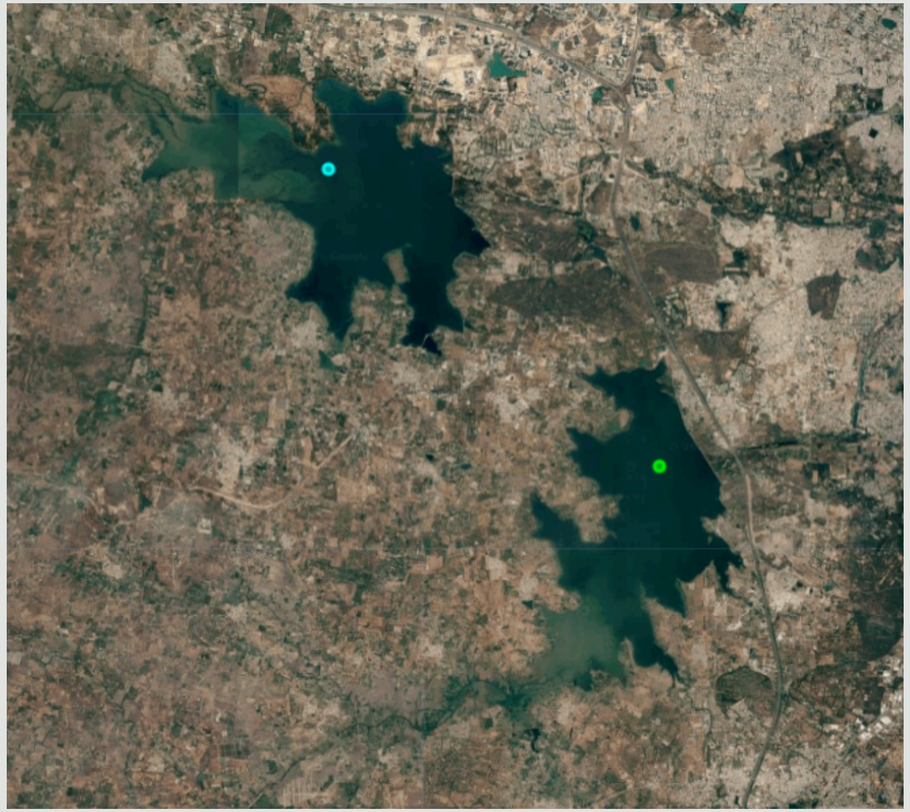
# AREA OF THE PROJECT

- This project is an interdisciplinary application of Artificial Intelligence and Remote Sensing for environmental monitoring.
- It focuses on analyzing temporal changes in the surface area of Hyderabad's urban lakes using satellite imagery and deep learning techniques.

- Remote Sensing & Geospatial Analysis
- Artificial Intelligence & Deep Learning
- Computer Vision
- Environmental Monitoring
- Urban Water Resource Management

## Study Lakes

- Hussain Sagar
- Osman Sagar
- Himayat Sagar



Satellite view of Hyderabad showing selected urban lakes.

# BACKGROUND OF THE PROBLEM



Aerial view of Hussain Sagar Lake, Hyderabad.

- Urban lakes support groundwater recharge, flood control, and biodiversity.
- Urbanization, pollution, and encroachment are degrading Hyderabad's lakes.
- Monitoring lake surface area is essential for environmental conservation.
- Traditional monitoring is manual and time-consuming.
- Satellite imagery + Deep Learning enables efficient long-term monitoring.

# PROBLEM STATEMENT

- Hyderabad's urban lakes are undergoing changes due to urbanization, pollution, and encroachment.
- Existing monitoring methods are manual, time-consuming, and difficult to scale.
- There is a need for an automated and reliable approach to monitor lake surface area over time.

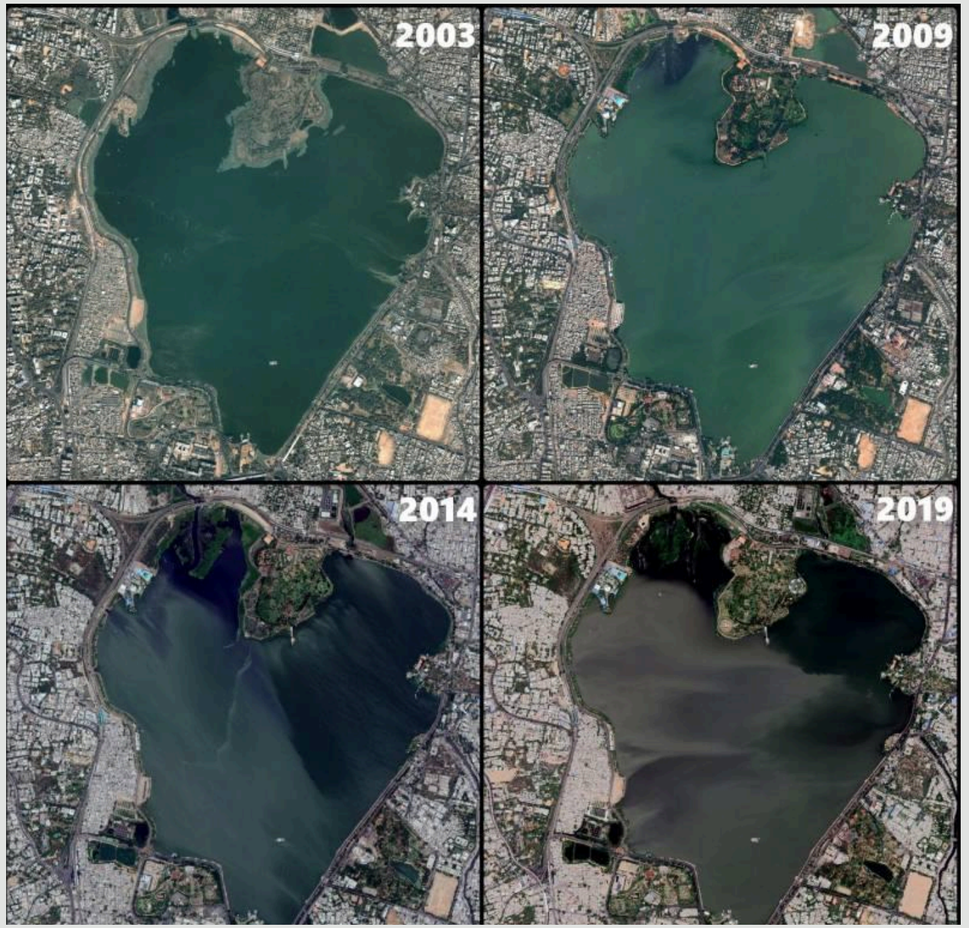



Figure: Multi-temporal satellite images illustrating changes in Hussain Sagar Lake.

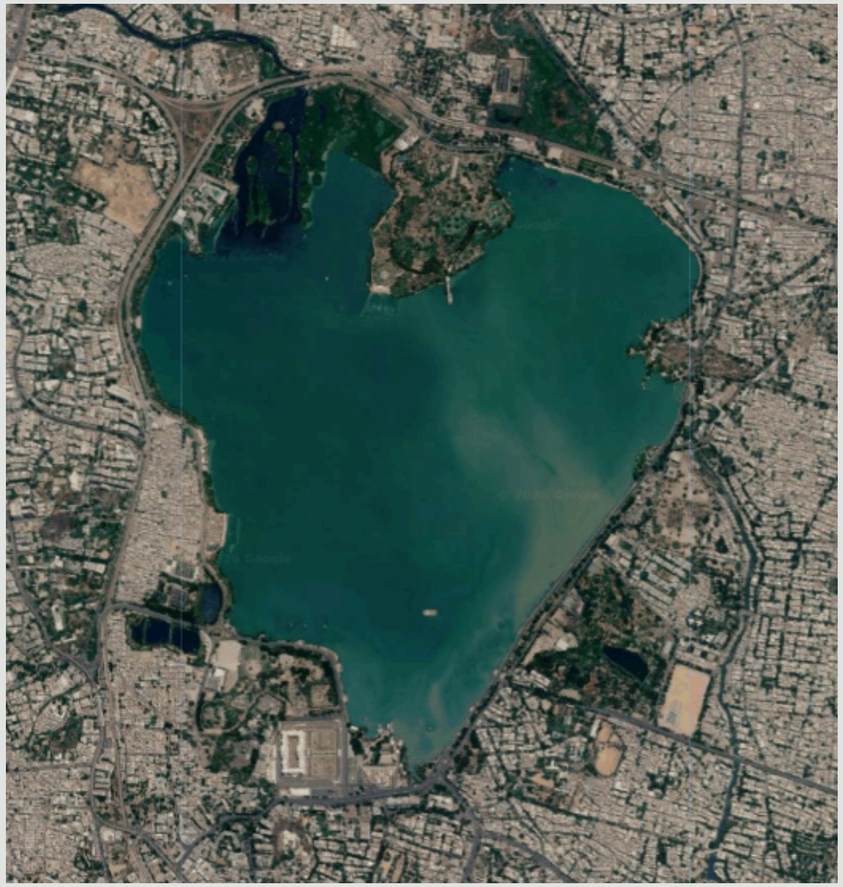
# PROPOSED WORK

- Collect Sentinel-2 satellite imagery (2018–2025).
- Preprocess satellite images for analysis.
- Apply U-Net-based semantic segmentation to identify water bodies.
- Estimate year-wise lake surface area and analyze temporal changes.

# NOVELTY

- Focuses on Hyderabad's three major urban lakes.
- Integrates Sentinel-2 imagery with U-Net-based segmentation.
- Enables automated monitoring of water bodies.
- Provides year-wise quantitative analysis of lake surface area changes.

# CONCLUSION



Aerial view of Hussain Sagar Lake, Hyderabad.

Presented the need for automated monitoring of Hyderabad's urban lakes.

Selected Sentinel-2 imagery and Hyderabad's three major lakes for the study.

Proposed a U-Net-based deep learning framework for water body segmentation.

Future work includes model development, surface area estimation, and temporal analysis.

# THANK YOU

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